

Anindya Mondal

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Education

Surrey Institute for People-centred AI, CVSSP, University of Surrey <i>PhD Candidate in Artificial Intelligence</i> Research focus: Integrating auxiliary information for natural world understanding.	Guildford, United Kingdom <i>Oct. 2022 – present</i>
Jadavpur University <i>B.E. (Hons.) in Electronics & Telecommunication Engineering, GPA: 8.79/10</i>	Kolkata, India <i>Aug. 2018 – Jun. 2022</i>

Research Focus

2023–2025: Vision–language integration, LLM, multimodal learning, action recognition, text-to-image synthesis, object counting, 3D/4D generation.

2020–2022: Graph neural networks, time-series analysis, graph signal processing, neuromorphic vision, subspace learning.

Publications

Preprint

Anindya Mondal, Ayan Banerjee, Sauradip Nag, Josep Lladós, Xiatian Zhu, Anjan Dutta, “CountLoop: Iterative Agent Guided High Instance Image Generation”. <https://mondalanindya.github.io/CountLoop/>

Published

AAAI '25: **Anindya Mondal**, Sauradip Nag, Xiatian Zhu, Anjan Dutta, “OmniCount: Multi-label Object Counting with Semantic–Geometric Priors.” Proceedings of AAAI 2025. 10.1609/aaai.v39i18.34151

ICCVW '23: **Anindya Mondal**, Sauradip Nag, Joaquin M. Prada, Xiatian Zhu, Anjan Dutta, “Actor-Agnostic Multi-label Action Recognition with Multi-modal Query.” ICCV Workshops 2023. 10.1109/ICCVW60793.2023.00086

ICASSP '23: J. A. C. Correa, J. H. Giraldo, **Anindya Mondal et al.**, “Time-Varying Signals Recovery via Graph Neural Networks.” ICASSP 2023. 10.1109/ICASSP49357.2023.10096168

EUSIPCO '22: **Anindya Mondal et al.**, “Recovery of Missing Sensor Data by Reconstructing Time-Varying Graph Signals.” EUSIPCO 2022. 10.23919/EUSIPCO55093.2022.9909940

ICCVW '21: **Anindya Mondal**, R. Shashant et al., “Moving Object Detection for Event-based Vision using Graph Spectral Clustering.” ICCV Workshops 2021. 10.1109/ICCVW54120.2021.00103

Research Experience

Surrey Institute for People-centred AI, CVSSP, University of Surrey <i>Doctoral Researcher</i>	Guildford, United Kingdom <i>Oct. 2022 – present</i>
○ Developed an LLM-guided, training-free framework that equips diffusion models with precise instance control for high-quality image generation. ○ Created a class-agnostic object-counting model using semantic and geometric priors. ○ Designed a transformer-based multimodal action-recognition model, improving precision by 50%. ○ Led the creation of a benchmark for animal action recognition, detection, and segmentation.	

Indian Institute of Science <i>Research Intern</i>	Bengaluru, India <i>May 2022 – Aug. 2022</i>
○ Devised a source-free domain-adaptation framework for image classification, enhancing robustness.	

- Implemented a Sobolev-norm minimisation algorithm for time-varying graph-signal reconstruction.
- Developed a graph-based semi-supervised learning framework for semantic segmentation.
- Improved moving-object detection from event data using graph spectral clustering.

Technical Skills

Languages: Python (expert), MATLAB (expert), C (advanced)

Libraries: PyTorch (expert), scikit-learn (advanced), NumPy (advanced), SciPy (advanced), Pandas (advanced)

Tools: Git (expert), L^AT_EX (expert), Jupyter Notebook (expert), Docker (advanced)

Awards and Honours

2024: AAAI 2025 Conference & Travel Grant, Philadelphia, USA — conference travel grant.

2023: ICCV 2023 Conference Grant, Paris, France — conference travel grant.

2022: Postgraduate Studentship, University of Surrey, UK — merit-based scholarship for doctoral studies.

2022: Uplink Research Internship Award, ACM SIGKDD India Chapter — internship and travel grant.

Professional Experience

Teaching

2023 & 2025: Teaching Assistant — Applied Machine Learning (M-level), Advanced Topics in Computer Vision & Deep Learning, University of Surrey.

Peer Review

2022–2025: Reviewer — ECCV, NeurIPS, BMVC, IEEE ISBI, ICASSP, IEEE TSP, ICCV Workshops.